Homework 4

ECEN 5156 - Physical Optics

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1.) Null Polarimetry

The null in the transmission is found with $\theta_{QWP} = 30^\circ$ and $\theta_{LA} = 60^\circ$ with respect to x. The angle for which a null is found must have the incident light be perpendicular to the linear polarizer such that:

$$\theta_{LA,null} = 60^\circ + 90^\circ = 150^\circ$$

Working backwards to find the angle that this polarized light (after going through the QWP) must be with respect to the FA of the QWP we get:

$$\alpha = \theta_{LA,null} - \theta_{QWP} = 120^\circ$$

The Jones vector for this polarization in between the two elements can be described with:

$$E_{between} = \begin{pmatrix} \cos\alpha \\ \sin\alpha \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix}$$

We can then use the Jones matrix for the QWP to work backward to get the initial polarization E_0:

$$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix} E_{between} = \begin{pmatrix} -\frac{1}{2} \\ i\frac{\sqrt{3}}{2} \end{pmatrix}$$

The ellipticity can be defined as the ratio of the minor axis (b) and the major axis (a):

$$\epsilon = \frac{b}{a} = \left| \frac{\frac{-1}{2}}{i\frac{\sqrt{3}}{2}} \right| = \frac{1}{\sqrt{3}}$$

The ellipticity angle is found with:

$$\chi = \tan^{-1}\left|\frac{b}{a}\right| = 30^\circ$$

The orientation angle has been found when we rotated the frame to match the polarization and is:

$$\psi = \alpha = 120^\circ$$

2.) Optoisolator

2a.)

The system can be treated as a series of matrices representing the light propagating forward through the optical elements and then in reverse after it hits the mirror.

$$\underline{M}_{final} = \underline{M}_{HP,B}\underline{M}_{QWP,B}\underline{M}_{mirror}\underline{M}_{QWP,F}\underline{M}_{HP,F}$$

Assuming the light enters the system unpolarized, the first element it interacts with is the horizontal polarizer $(\underline{M}_{HP})$ with a matrix represented by:

$$\underline{M}_{HP} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

With the matrix for the quarter waveplate $(\underline{M}_{QWP})$ represented as:

$$\underline{M}_{QWP} = e^{i\pi/4}\begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}$$

But since it is rotated with its fast axis at $+45^\circ$ we must use the rotation matrices.

$$\underline{M}_{QWP,+45^\circ} = \underline{R}(-\theta)\underline{M}_{QWP}\underline{R}(\theta)$$

$$\underline{M}_{QWP,+45^\circ} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ -1 & 1\end{pmatrix} e^{i\pi/4}\begin{pmatrix}1 & 0\\ 0 & -i\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1 & -1\\ 1 & 1\end{pmatrix}$$

$$\underline{M}_{QWP,+45^\circ} = \frac{1}{2}e^{i\pi/4}\begin{pmatrix}1-i & 1+i\\ 1+i & 1-i\end{pmatrix}$$

$$\underline{M}_{QWP,+45^\circ} = \frac{1}{2}e^{i\pi/4}(1-i)\begin{pmatrix}1 & i\\ i & 1\end{pmatrix}$$

$$\underline{M}_{QWP,+45^\circ} = \frac{1}{2}\left[\frac{1+i}{\sqrt{2}}\right](1-i)\begin{pmatrix}1 & i\\ i & 1\end{pmatrix}$$

$$\underline{M}_{QWP,+45^\circ} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & i\\ i & 1\end{pmatrix}$$

For the mirror ($\underline{M}_{mirror}$), the matrix is:

$$\underline{M}_{mirror} = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$$

On the return trip the QWP is now at -45° which gives:

$$\underline{M}_{QWP,-45^\circ} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & -i\\ -i & 1\end{pmatrix}$$

Combining this all together:

$$\underline{M}_{final} = \begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1 & -i\\ -i & 1\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1 & i\\ i & 1\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}$$

```
m_qwp_b = [1 -im ; -im 1]
m_hp = [1 0 ; 0 0]
m_qwp_f = [1 im ; im 1]
m_mirror = [1 0 ; 0 -1]

m_f = m_hp * (1/sqrt(2)) * m_qwp_b * m_mirror * (1/sqrt(2)) * m_qwp_f * m_hp

m_before = (1/sqrt(2)) * m_qwp_b * m_mirror * (1/sqrt(2)) * m_qwp_f * m_hp

println(m_f)
println(m_before)
```

```
ComplexF64[0.0 + 0.0im 0.0 + 0.0im; 0.0 + 0.0im 0.0 + 0.0im]
ComplexF64[0.0 + 0.0im 0.0 + 0.0im; 0.0 - 0.9999999999999998im 0.0 + 0.0im]
```

As seen above, the light does not make it through the entire system as right before it reaches the horizontal polarizer again it is vertically polarized.

2b.)

The general form for a phase retarder is:

$$\underline{M} = \begin{pmatrix}e^{i\delta/2} & 0\\ 0 & e^{-i\delta/2}\end{pmatrix}$$

If the phase is defined to be $\delta = 90 + \Delta$, the matrix can be rewritten as:

$$\underline{M} = \begin{pmatrix}e^{i\Delta/2}e^{i\pi/4} & 0\\ 0 & e^{-i\Delta/2}e^{-i\pi/4}\end{pmatrix}$$

$$= e^{i\pi/4}\begin{pmatrix}e^{i\Delta/2} & 0\\ 0 & -ie^{-i\Delta/2}\end{pmatrix}$$

$$= e^{i\pi/4}\begin{pmatrix}\cos(\frac{\Delta}{2}) + i\sin(\frac{\Delta}{2}) & 0\\ 0 & -i(\cos(\frac{\Delta}{2}) - i\sin(\frac{\Delta}{2}))\end{pmatrix}$$

Assuming small Δ with $\cos(\frac{\Delta}{2}) \approx 1$ and $\sin(\frac{\Delta}{2}) \approx \frac{\Delta}{2}$

$$\approx e^{i\pi/4}\begin{pmatrix}1 + i\frac{\Delta}{2} & 0\\ 0 & -i - \frac{\Delta}{2}\end{pmatrix}$$

$$\approx e^{i\pi/4}\left[\begin{pmatrix}1 & 0\\ 0 & -i\end{pmatrix}+\frac{\Delta}{2}\begin{pmatrix}i & 0\\ 0 & -1\end{pmatrix}\right]$$

$$\approx \underline{M}_{QWP+45^\circ}+e^{i\pi/4}\frac{\Delta}{2}\begin{pmatrix}i & 0\\ 0 & -1\end{pmatrix}$$

Since I was able to seperate the standard QWP matrix from the deviated one, I can treat this problem as a perturbation with:

$$\underline{M}_{final}=\underline{M}^{(0)}+\underline{M}^{(1)}$$

The QWP is the one with the perturbation so we rewrite it as:

$$\underline{M}_{final}=\underline{M}_{HP,B}\left[(\underline{M}^{(0)}_{QWP,B}+\underline{M}^{(1)}_{QWP,B})\underline{M}_{mirror}(\underline{M}^{(0)}_{QWP,F}+\underline{M}^{(1)}_{QWP,F})\right]\underline{M}_{HP,F}$$

$$\underline{M}_{final}=\underline{M}_{HP,B}\left[(\underline{M}^{(0)}_{QWP,B}\underline{M}_{mirror}+\underline{M}_{mirror}\underline{M}^{(1)}_{QWP,B})(\underline{M}^{(0)}_{QWP,F}+\underline{M}^{(1)}_{QWP,F})\right]\underline{M}_{HP,F}$$

$$\underline{M}_{final}=\underline{M}_{HP,B}\left[(\underline{M}^{(0)}_{QWP,B}\underline{M}_{mirror}\underline{M}^{(0)}_{QWP,F})+(\underline{M}^{(0)}_{QWP,B}\underline{M}_{mirror}\underline{M}^{(1)}_{QWP,F}+\underline{M}^{(1)}_{QWP,B}\underline{M}_{mirror}\underline{M}^{(0)}_{QWP,F})+(\underline{M}^{(1)}_{QWP,B}\underline{M}_{mirror}\underline{M}^{(1)}_{QWP,F})\right]\underline{M}$$

The first term in the brackets can be removed as it is the unperturbed path and results in zero.

$$\underline{M}_{final}=\underline{M}_{HP,B}\left[(\underline{M}^{(0)}_{QWP,B}\underline{M}_{mirror}\underline{M}^{(1)}_{QWP,F}+\underline{M}^{(1)}_{QWP,B}\underline{M}_{mirror}\underline{M}^{(0)}_{QWP,F})+(\underline{M}^{(1)}_{QWP,B}\underline{M}_{mirror}\underline{M}^{(1)}_{QWP,F})\right]\underline{M}_{HP,F}$$

The matrices of the cross terms are calculated to be:

$$\underline{M}^{(0)}_{QWP,B}\underline{M}_{mirror}\underline{M}^{(1)}_{QWP,F}=\frac{\Delta}{2}\begin{pmatrix}-1 & 0\\ 0 & 0\end{pmatrix}$$

$$\underline{M}^{(1)}_{QWP,B}\underline{M}_{mirror}\underline{M}^{(0)}_{QWP,F}=\frac{\Delta}{2}\begin{pmatrix}-1 & 0\\ 0 & 0\end{pmatrix}$$

While the perturbation terms only results in zero:

$$\underline{M}^{(1)}_{QWP,B}\underline{M}_{mirror}\underline{M}^{(1)}_{QWP,F}=\frac{\Delta^2}{4}\begin{pmatrix}0 & 0\\ 0 & 0\end{pmatrix}=0$$

The resulting value for the matrix is thus:

$$\underline{M}_{final}=\frac{\Delta}{2}\begin{pmatrix}-2 & 0\\ 0 & 0\end{pmatrix}$$

Which we solve for the resulting x component of the electric field to be:

$$E_x=-\Delta$$

Where we can find the power as:

$$P=|-\Delta|^2=\Delta^2$$

```
m_qwp_b_pert = [-1 -im ; -im -1]
m_hp = [1 0 ; 0 0]
m_qwp_f_pert = [-1 im ; im -1]
m_mirror = [1 0 ; 0 -1]

m_f_cross_1 = m_hp * (1/sqrt(2)) * m_qwp_b * m_mirror * (1/sqrt(2)) * m_qwp_f_pert * m_hp
m_f_cross_2 = m_hp * (1/sqrt(2)) * m_qwp_b_pert * m_mirror * (1/sqrt(2)) * m_qwp_f * m_hp
m_f_pert_only = m_hp * (1/sqrt(2)) * m_qwp_b_pert * m_mirror * (1/sqrt(2)) * m_qwp_f_pert * m_hp


println(m_f_cross_1)
println(m_f_cross_2)
println(m_f_pert_only)
```

```
ComplexF64[-0.9999999999999998 + 0.0im 0.0 + 0.0im; 0.0 + 0.0im 0.0 + 0.0im
]
ComplexF64[-0.9999999999999998 + 0.0im 0.0 + 0.0im; 0.0 + 0.0im 0.0 + 0.0im
]
ComplexF64[0.0 + 0.0im 0.0 + 0.0im; 0.0 + 0.0im 0.0 + 0.0im]
```

3.) Jones to Orientation

The E_x and E_y components of the electric field are:

$$E_x = E_{0,x} e^{i\phi_x}$$

$$E_y = E_{0,x} e^{i\phi_y}$$

With the phase difference being given as:

$$\delta = \phi_y - \phi_x$$

The ratio of the electric fields can be represented with respect to an auxilliary angle α:

$$\tan\alpha = \frac{E_{0,y}}{E_{0,x}}$$

The relation to the orientation of the ellipse is given by:

$$\tan 2\Psi = \tan 2\alpha \cos\delta$$

Using the trig identity:

$$\tan 2\alpha = \frac{2\tan\alpha}{1-\tan^2\alpha}$$

$$= \frac{2\frac{E_{0,y}}{E_{0,x}}}{1-(\frac{E_{0,y}}{E_{0,x}})^2}$$

$$= \frac{2E_{0,y}E_{0,x}}{E_{0,x}^2 - E_{0,y}^2}$$

Plugging this back in we arrive at the desired relation:

$$\tan 2\Psi = \frac{2E_{0,y}E_{0,x}}{E_{0,x}^2 - E_{0,y}^2}\cos\delta$$

Now for the ellipticity angle (χ) we use a similar process and then relate it to $(\backslash\text{Chi})$:

$$\sin 2\chi = \sin 2\alpha \sin\delta$$

Using the trig identity:

$$\sin 2\alpha = \frac{2\tan\alpha}{1+\tan^2\alpha}$$

$$= \frac{2\frac{E_{0,y}}{E_{0,x}}}{1+(\frac{E_{0,y}}{E_{0,x}})^2}$$

$$= \frac{2E_{0,y}E_{0,x}}{E_{0,x}^2 + E_{0,y}^2}$$

$$\sin 2\chi = \frac{2E_{0,y}E_{0,x}}{E_{0,x}^2 + E_{0,y}^2}\sin\delta$$

$$\backslash\text{Chi}$$

is defined to be:

$$\tan\backslash\text{Chi} = \frac{a}{b}$$

While χ can be related with:

$$\tan\chi = \frac{b}{a}$$

Therefore:

$$\tan\backslash\text{Chi} = \cot\chi$$

$$\tan\backslash\text{Chi} = \tan(\frac{\pi}{2} - \chi)$$

$$\backslash\text{Chi} = \frac{\pi}{2} - \chi$$

$$2\backslash\text{Chi} = \pi - 2\chi$$

$$\sin 2\backslash\text{Chi} = \sin(\pi - 2\chi)$$

$$\sin 2\Chi = \sin(2\chi)$$

Therefore:

$$\sin 2\Chi = \frac{2E_{0,y}E_{0,x}}{E_{0,x}^2 + E_{0,y}^2}\sin\delta$$

4.) Circular basis polarization decomposition

The superposition can be represented as:

$$\vec{J} = E_L\vec{J}_L + E_R\vec{J}_R$$

$$= E_L\frac{1}{\sqrt{2}}\begin{bmatrix}1\\ i\end{bmatrix} + E_R\frac{1}{\sqrt{2}}\begin{bmatrix}1\\ -i\end{bmatrix}$$

$$= E_{0,L}e^{i\delta_L}\frac{1}{\sqrt{2}}\begin{bmatrix}1\\ i\end{bmatrix} + E_{0,R}e^{i\delta_R}\frac{1}{\sqrt{2}}\begin{bmatrix}1\\ -i\end{bmatrix}$$

The orientation angle is found with:

$$\psi = \frac{1}{2}(\delta_L - \delta_R)$$

And the ellipticity angle χ can be found with:

$$\chi = \tan^{-1}\left(\frac{b}{a}\right)$$

The major axis is represented by the sum of the magnitudes while the minor axis is the difference:

$$a = E_{0,L} + E_{0,R}$$

$$b = |E_{0,L} - E_{0,R}|$$

$$\tan\chi = \frac{E_{0,L} - E_{0,R}}{E_{0,L} + E_{0,R}}$$

$$\tan\chi = \frac{1 - \frac{E_{0,R}}{E_{0,L}}}{1 + \frac{E_{0,R}}{E_{0,L}}}$$

To get it in terms of an auxilliary angle again we can do:

$$\tan(\frac{\pi}{4} - \alpha) = \frac{\tan(\frac{\pi}{4}) - \tan\alpha}{1 + \tan(\frac{\pi}{4})\tan\alpha} = \frac{1 - \tan\alpha}{1 + \tan\alpha}$$

$$\tan\alpha = \frac{E_{0,R}}{E_{0,L}} = \frac{\sin\alpha}{\cos\alpha}$$

$$\chi = \frac{\pi}{4} - \alpha$$

$$\alpha = \frac{\pi}{4} - \chi$$

$$E_{0,R} = E_0\sin(\frac{\pi}{4} - \chi)$$

$$E_{0,L} = E_0\cos(\frac{\pi}{4} - \chi)$$

$$\vec{J} = E_0\cos(\frac{\pi}{4} - \chi)e^{i\delta_L}\frac{1}{\sqrt{2}}\begin{bmatrix}1\\ i\end{bmatrix} + E_0\sin(\frac{\pi}{4} - \chi)e^{i\delta_R}\frac{1}{\sqrt{2}}\begin{bmatrix}1\\ -i\end{bmatrix}$$

$$\vec{J} = E_0\cos(\frac{\pi}{4} - \chi)e^{i(2\psi+\delta_R)}\frac{1}{\sqrt{2}}\begin{bmatrix}1\\ i\end{bmatrix} + E_0\sin(\frac{\pi}{4} - \chi)e^{i(\delta_L - 2\psi)}\frac{1}{\sqrt{2}}\begin{bmatrix}1\\ -i\end{bmatrix}$$

Setting δ_R to zero arbitrarily as we are working in relative phases:

$$\vec{J} = E_0\cos(\frac{\pi}{4} - \chi)e^{i2\psi}\frac{1}{\sqrt{2}}\begin{bmatrix}1\\ i\end{bmatrix} + E_0\sin(\frac{\pi}{4} - \chi)\frac{1}{\sqrt{2}}\begin{bmatrix}1\\ -i\end{bmatrix}$$

$$\vec{J} = \frac{E_0}{\sqrt{2}}\begin{bmatrix}\cos(\frac{\pi}{4} - \chi)e^{i2\psi} + \sin(\frac{\pi}{4} - \chi)\\ i\cos(\frac{\pi}{4} - \chi)e^{i2\psi} - i\sin(\frac{\pi}{4} - \chi)\end{bmatrix}$$

```
using Plots
plotly()

E_0 = 1
χ = pi/6
ψ = pi/6

# J = (E_0 / sqrt(2)) * [cos((pi/4) - χ) * exp(im*2(ψ)) + sin((pi/4) - χ) ; im*cos((pi/4) - χ) * exp(im*2(ψ)) - im*sin((pi/4)

E_x_complex = (E_0 / sqrt(2)) * (cos((pi/4) - χ) + sin((pi/4) - χ)* exp(im*2*(ψ)))
E_y_complex = (E_0 / sqrt(2)) * (im*cos((pi/4) - χ)  - im*sin((pi/4) - χ)* exp(im*2*(ψ)))

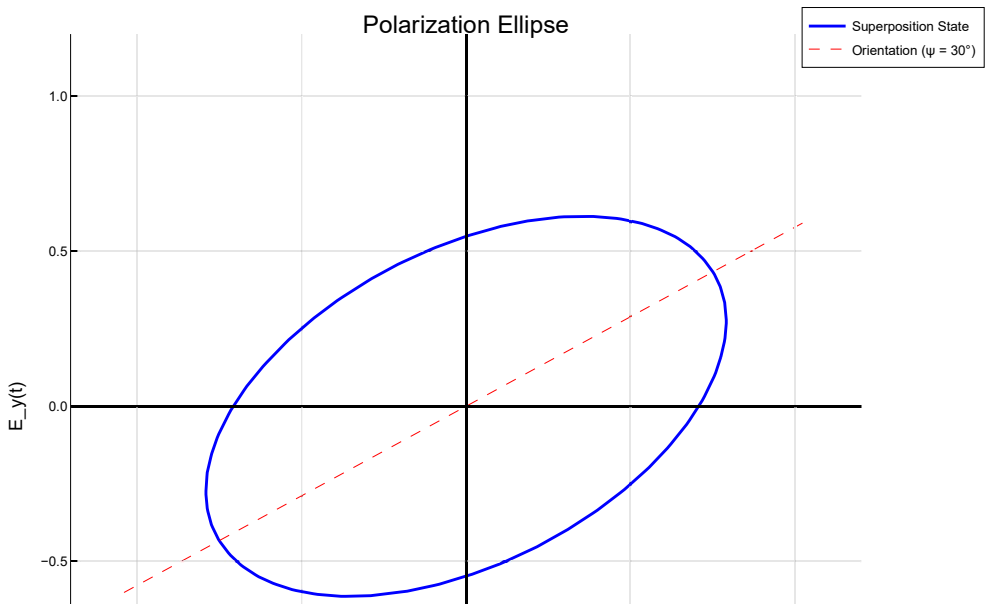
ωt = range(0, 2π, length=300)

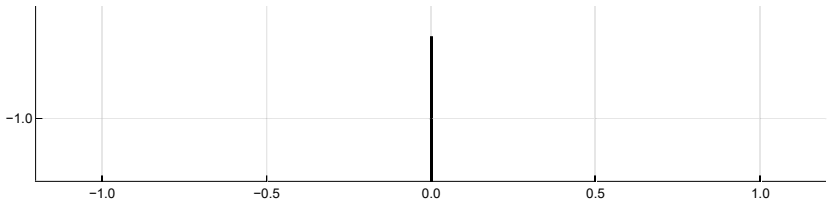
E_x_real = real.(E_x_complex .* exp.(-im .* ωt))
E_y_real = real.(E_y_complex .* exp.(-im .* ωt))

plot(
    E_x_real,
    E_y_real,
    label="Superposition State",
    linewidth=2.5,
    color=:blue,
    aspect_ratio=:equal,
    xlabel="E_x(t)",
    ylabel="E_y(t)",
    title="Polarization Ellipse",
    xlims=(-1.2, 1.2),
    ylims=(-1.2, 1.2),
    size = (900, 700),
)

plot!(
    [-1.2*cos(ψ), 1.2*cos(ψ)],
    [-1.2*sin(ψ), 1.2*sin(ψ)],
    label="Orientation (ψ = 30°)",
    linestyle=:dash,
    color=:red
)

hline!(
    [0],
    [0],
    label="",
    color=:black,
    lw=2
)
vline!(
    [0],
    [0],
    label="",
    color=:black,
    lw=2
)
```





E_x(t)

```
using Plots
plotly()

E_0 = 1
χ = pi/6
ψ = pi/6

S_1 = cos(2*χ)*cos(2*ψ)
S_2 = cos(2*χ)*sin(2*ψ)
S_3 = sin(2*χ)
S_0 = sqrt(S_1^2 + S_2^2 + S_3^2)

scatter3d(
    [S_1],
    [S_2],
    [S_3],
    label = "(S1: $(round(S_1, digits=3)) \n S2: $(round(S_2, digits=3)) \n S3: $(round(S_3, digits=3)))",
    color = :red,
    markersize = 2,
    xlabel = "S1",
    ylabel = "S2",
    zlabel = "S3",
    title = "Poincaré Sphere",
    xlims = (-1, 1),
    ylims = (-1, 1),
    zlims = (-1, 1),
    size = (900, 700),
)

axis_lw = 3
axis_color = :black


plot!(
    [-1, 1],
    [0, 0],
    [0, 0],
    color=axis_color,
    linewidth=axis_lw,
    label=""
)

plot!(
    [0, 0],
    [-1, 1],
    [0, 0],
    color=axis_color,
    linewidth=axis_lw,
    label=""
)

plot!(
    [0, 0],
    [0, 0],
    [-1, 1],
    color=axis_color,
    linewidth=axis_lw,
    label=""
)

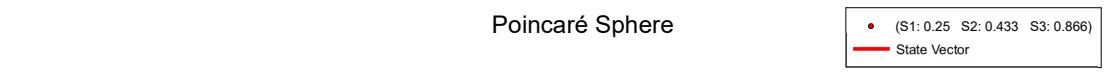
u = range(0, 2π, length=100)
v = range(0, π, length=100)

xs = [cos(ui)*sin(vi) for ui in u, vi in v]
ys = [sin(ui)*sin(vi) for ui in u, vi in v]
zs = [cos(vi) for ui in u, vi in v]


wireframe!(
    xs,
    ys,
```

```
    zs,
    color = :gray,
    alpha = 0.85,
)

plot!(
    [0, S_1],
    [0, S_2],
    [0, S_3],
    color = :red,
    linewidth = 3,
    label = "State Vector"
)
```



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